

Advanced (Habanero)

- Q1.** Which of the following is a condition for a function to have an inverse?
- (A) The function must be one-to-one
(B) The function must be many-to-one
(C) The function must be a linear
(D) The function must be a quadratic
(E) The function must be a polynomial
- Q2.** Find the inverse of the function $f(x) = 2x + 1$
- (A) $f^{-1}(x) = \frac{x-1}{2}$
(B) $f^{-1}(x) = \frac{x+1}{2}$
(C) $f^{-1}(x) = 2x - 1$
(D) $f^{-1}(x) = 2x + 1$
(E) $f^{-1}(x) = \frac{x}{2} - 1$
- Q3.** If $f(x) = x^2$ and $g(x) = \sqrt{x}$, which statement is true?
- (A) $f(g(x)) = x$
(B) $g(f(x)) = x$
(C) $f(g(x)) = x^2$
(D) $g(f(x)) = \sqrt{x}$
(E) $f(g(x)) = \sqrt{x^2}$
- Q4.** Which function is the inverse of $g(x) = \frac{1}{x}$?
- (A) $f(x) = \frac{1}{x}$
(B) $f(x) = x$
(C) $f(x) = x^2$
(D) $f(x) = \frac{1}{x^2}$
(E) $f(x) = \frac{x}{x}$
- Q5.** If $f(x) = x^3 + 1$, find $f^{-1}(x)$
- (A) $f^{-1}(x) = \sqrt[3]{x-1}$
(B) $f^{-1}(x) = \sqrt[3]{x+1}$
(C) $f^{-1}(x) = x^3 - 1$
(D) $f^{-1}(x) = x^3 + 1$
(E) $f^{-1}(x) = \sqrt[3]{x}$
- Q6.** Which statement is true for the inverse function $f^{-1}(x)$?
- (A) $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$
(B) $f(f(x)) = x$ and $f^{-1}(f^{-1}(x)) = x$
(C) $f(f(x)) = x$ and $f^{-1}(f^{-1}(x)) = x$
(D) $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = f(x)$
(E) $f(f^{-1}(x)) = f(x)$ and $f^{-1}(f(x)) = x$
- Q7.** The function $f(x) = x^2 - 4$ has an inverse if its domain is restricted to
- (A) $x \geq 0$
(B) $x \leq 0$
(C) $x > 0$
(D) $x < 0$
(E) $x \geq 2$
- Q8.** If $f(x) = \frac{1}{x-1}$, find the inverse function $f^{-1}(x)$
- (A) $f^{-1}(x) = \frac{x+1}{x}$
(B) $f^{-1}(x) = \frac{x-1}{x}$
(C) $f^{-1}(x) = \frac{1}{x} + 1$
(D) $f^{-1}(x) = \frac{1}{x} - 1$
(E) $f^{-1}(x) = \frac{1}{x-1}$

Open Ended Questions (FRQ)

- Q9.** Show that the function $f(x) = x^3$ has an inverse and find $f^{-1}(x)$. Verify your answer using composition.
- Q10.** A company's profit P is given by the function $P(x) = 200x - 0.5x^2$, where x is the number of units sold. Find the inverse function $P^{-1}(x)$ and interpret it in the context of the problem.

Chef's Tips

- Remind students to switch x and y when finding the inverse of a function.
- Caution students to restrict the domain of a function when necessary to ensure it has an inverse.
- Encourage students to verify their answers using composition.